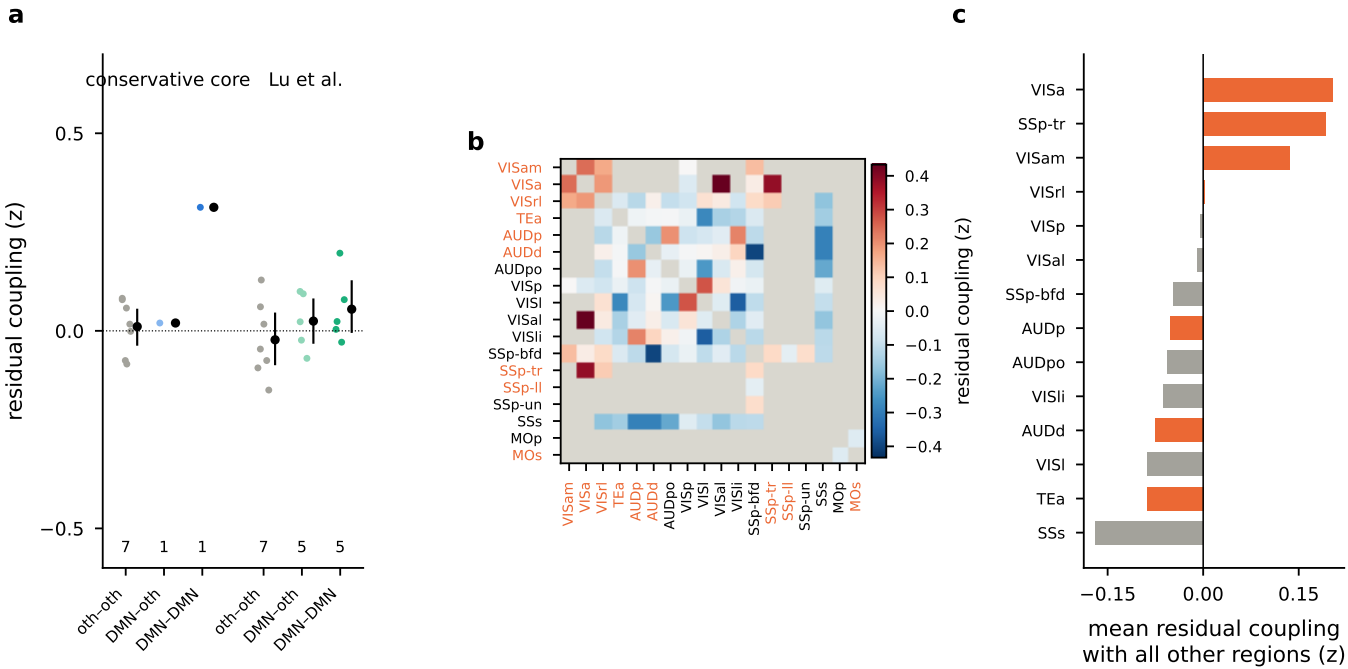

Figure 7: a Mesoscope grey-screen period: sessions with ≥200 s (most canonical sessions have only 60 s) and motion energy to verify stillness, 13 sessions, 7 mice (all 23 long sessions, without the motion-energy requirement, give the same ordering). Region-averaged calcium activity in still 1 s bins with nuisance regression; coupling as residual from the distance-matched mean; dots are mice, black mean and 95% mouse-bootstrap CI; numbers at the bottom give the mice per class (the mesoscope images only the RSP part of the conservative core, giving a single core-core pair, RSPagl-RSPd, from one mouse). No frontal DMN region is imaged in these sessions; the wheel is logged in only 8 of the 23 long sessions. **b** Residual coupling of every imaged region pair present in ≥2 mice (grey: pair not imaged in ≥2 mice; regions without any such pair omitted), ordered posterior DMN, Lu temporal (TEa, AUD), visual, somatosensory, motor; tick colours as in Fig. 4. **c** Mean residual coupling of each region with all other imaged regions (pairs in ≥2 mice). VISa and VISam are the most coupled regions together with trunk somatosensory cortex (SSp-tr); RSP has too few pairs in this subset; the auditory areas and TEa (Lu et al. temporal module) are among the least coupled.